Assignment 5 -solns

1. In the ground state of hydrogen, what is the probability that the electron is found inside the Bohr radius?

Ans. We can work with just one coordinate, the radial distance from the nucleus, r.

In atomic units, the ground state wavefunction is $R = 2e^{-r}$and Bohr radius= 1unit of distance.

$\therefore$ the probability that the electron is found inside the Bohr radius

$= \int_0^1 r^2 (2e^{-r})^2 dr = 4\int_0^1 r^2 e^{-2r} dr = 4\left[-r^2e^{-2r} - \frac{1}{2}re^{-2r} - \frac{1}{4}e^{-2r}\right]\Big|_0^1 = -5e^{-2} + 1 \approx 0.323$

2. Find the bound state energy levels for a particle of mass μ moving in the central potential:

$$V(r) = -\frac{A}{r} + \frac{B}{r^2}$$

Ans. let $A = Ze^2$; $\quad B = \frac{\hbar^2}{2\mu}\gamma$

Recall : For the hydrogenic case, the radial equation is : $\frac{d^2u}{dr^2} + \frac{2\mu E}{\hbar^2}u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2}\right]u$

Thus the radial equation for a given l for the given problem is is exactly as for the hydrogenic case, but with $l(l+1)$ replaced by $l(l+1) + \gamma$

Define: $l' = -\frac{1}{2} + \sqrt{\left(l+\frac{1}{2}\right)^2 + \gamma}$ so that

$l(l+1) + \gamma = l'(l'+1)$ and the radial eq. is as in hydrogen with $l \to l'$.

with energy $E = -\epsilon$ and $\rho_0 = \frac{Ze^2}{2\pi\epsilon_0\hbar}\sqrt{\frac{\mu}{2\epsilon}}$

allowed values of ρ_0 are: $\rho_0 = 2(N + l')$

Let $n = N + l' = 1, 2, \cdots$, and $l' = 0,\ 1,\ 2\cdots,\ n-1$

$\implies N = n - l'$

$\therefore \rho_0 = 2\left[n - \left(l+\frac{1}{2}\right) + \sqrt{\left(l+\frac{1}{2}\right)^2 + \gamma}\right]$

and $E_{n,l} = -\frac{\left(Ze^2\right)^2}{2\hbar^2}\mu\frac{1}{\left[n-\left(l+\frac{1}{2}\right)+\sqrt{\left(l+\frac{1}{2}\right)^2+\gamma}\right]^2} \equiv -\frac{A^2}{2\hbar^2}\mu\frac{1}{\left[n-\left(l+\frac{1}{2}\right)+\sqrt{\left(l+\frac{1}{2}\right)^2+\frac{2\mu}{\hbar^2}B}\right]^2}$

3. (i) Consider the $n = 2$ states of a hydrogenic atom. For a perturbation $H' = \lambda\frac{\vec{\sigma}.\vec{r}}{r}$, verify that $H = H_0 + H'$ commutes with J^2 and J_z.

(ii) Consider the effect on a hydrogenic atom of a (hypothetical) parity-violating interaction $H' = \lambda\frac{\vec{\sigma}.\vec{r}}{r}$, where $\vec{\sigma}$ is a vector formed using the Pauli spin matrices.

Calculate the $n = 2$ energy levels (write the term symbols for the unperturbed states) to first order in H' and give the degeneracy of each level.

Ans. $H = H_0 + H'$ commutes with $J^2$and J_z, so the eigenstates can be taken to have definite quantum numbers j and M.

H' has odd parity, so it has zero matrix elements between states of the same parity (states of the same l).

For each M, let $S_{\frac{1}{2}}, P_{\frac{1}{2}}, P_{\frac{3}{2}}$ denote the unperturbed $n = 2$ wavefunctions, with the quantum numbers L and j implied by the symbols.

Since we can work separately for each $(j,\, M)$ and it is clear that the $P_{\frac{3}{2}}$ levels can be treated in isolation from $S_{\frac{1}{2}}$

(i) But $\langle P_{\frac{3}{2},\,M}|H'|P_{\frac{3}{2},\,M}\rangle = 0; \quad M = -\frac{3}{2}, ..., \frac{3}{2} \quad \Longrightarrow$ there is no first order shift for the $P_{\frac{3}{2}}$ level, so they remain at energy $E_2^{(0)}$ with 4-fold degeneracy (in M)

(ii) The $S_{\frac{1}{2}}$ and $P_{\frac{1}{2}}$ levels are connected by H' so have to be treated by degenerate perturbation methods. Clearly

$$\langle S_{\frac{1}{2},\,M}|H'|S_{\frac{1}{2},\,M}\rangle = \langle P_{\frac{1}{2},\,M}|H'|P_{\frac{1}{2},\,M}\rangle = 0$$

Thus we need only

$$\langle P_{\frac{1}{2},\,M}|H'|S_{\frac{1}{2},\,M}\rangle = \langle S_{\frac{1}{2},\,M}|H'|P_{\frac{1}{2},\,M}\rangle^* = A$$

and it is clear that A is the same for $M = \pm\frac{1}{2}$

moreover let us choose a phase convention : A is real.

For the $j = \frac{1}{2}$ vector the first order shifts are given by

$$\det \begin{vmatrix} -E^{(1)} & A \\ A & -E^{(1)} \end{vmatrix} = 0$$

so the energies to first order are $E_2^{(0)} + A$ or $E_2^{(0)} - A$, each being 2-fold degenerate ($M = \pm\frac{1}{2}$),

(iii) there remains to compute A. Since it doesn't depend on M for definiteness, take $M = \frac{1}{2}$.

Then $|S_{\frac{1}{2},\,M=\frac{1}{2}}\rangle = R_{20}(r)Y_0^0\chi^{\frac{1}{2}} = \frac{1}{\sqrt{4\pi}}R_{20}(r)\chi^{\frac{1}{2}}$ (note that this is the 2s atomic orbital)

But

$$\frac{\vec{\sigma}.\vec{r}}{r} = \begin{pmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{pmatrix}; \; \chi^{\frac{1}{2}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \chi^{-\frac{1}{2}} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

We can write it in a more convenient form. Before that, let us write $|P_{\frac{1}{2},\,\frac{1}{2}}\rangle = R_{21}(r)\left\{c_1Y_1^0\chi^{\frac{1}{2}} + c_2Y_1^1\chi^{-\frac{1}{2}}\right\}$ (note that for $M = \frac{1}{2}$, I have only two combinations, $Y_1^0\chi^{\frac{1}{2}}$ and $Y_1^1\chi^{-\frac{1}{2}}$,but we have to find the coefficients $c_1$and c_2).

Start with $|P_{\frac{3}{2},\,\frac{3}{2}}\rangle \sim Y_1^1\chi^{\frac{1}{2}}$ (ignore the radial part for now)

Apply $\widehat{J_-}|P_{\frac{3}{2},\,\frac{3}{2}}\rangle \sim \left(\widehat{L_-} + \widehat{S_-}\right)Y_1^1\chi^{\frac{1}{2}}$

or, $\hbar\sqrt{\left(\frac{3}{2} - \frac{3}{2} + 1\right)\left(\frac{3}{2} + \frac{3}{2}\right)}|P_{\frac{3}{2},\,\frac{1}{2}}\rangle \sim \hbar\sqrt{(1 - 1 + 1)(1 + 1)}Y_1^0\chi^{\frac{1}{2}} + \hbar\sqrt{\left(\frac{1}{2} - \frac{1}{2} + 1\right)\left(\frac{1}{2} + \frac{1}{2}\right)}Y_1^{-1}\chi^{-\frac{1}{2}}$

or, $|P_{\frac{3}{2},\,\frac{1}{2}}\rangle \sim \frac{\sqrt{2}}{\sqrt{3}}Y_1^0\chi^{\frac{1}{2}} + \frac{1}{\sqrt{3}}Y_1^{-1}\chi^{-\frac{1}{2}}$

Now, $|P_{\frac{1}{2},\,\frac{1}{2}}\rangle \sim \left\{c_1Y_1^0\chi^{\frac{1}{2}} + c_2Y_1^1\chi^{-\frac{1}{2}}\right\}$

using orthogonality $\langle P_{\frac{1}{2},\,\frac{1}{2}}|P_{\frac{3}{2},\,\frac{1}{2}}\rangle = 0$, obtain $c_1 = -\frac{1}{\sqrt{3}}; c_2 = \frac{\sqrt{2}}{\sqrt{3}}$

Thus, let us write $\frac{\vec{\sigma}.\vec{r}}{r}\langle S_{\frac{1}{2},\,M=\frac{1}{2}}|$ in a more convenient form :

$$\therefore \frac{\vec{\sigma}.\vec{r}}{r}\langle S_{\frac{1}{2},\,M=\frac{1}{2}}| = \frac{1}{\sqrt{4\pi}}R_{20}(r)\left\{\cos\theta\chi^{\frac{1}{2}} - \sin\theta e^{-i\phi}\chi^{-\frac{1}{2}}\right\} = R_{20}(r)\left\{\frac{1}{\sqrt{3}}Y_1^0\chi^{\frac{1}{2}} - \frac{\sqrt{2}}{\sqrt{3}}Y_1^{-1}\chi^{-\frac{1}{2}}\right\}$$

But

$$|P_{\frac{1}{2},\,\frac{1}{2}}\rangle = R_{21}(r)\left\{-\frac{1}{\sqrt{3}}Y_1^0\chi^{\frac{1}{2}} + \frac{\sqrt{2}}{\sqrt{3}}Y_1^1\chi^{-\frac{1}{2}}\right\}$$

$$\therefore A = \lambda\langle P_{\frac{1}{2},\,\frac{1}{2}}|H'|S_{\frac{1}{2},\,\frac{1}{2}}\rangle = -\lambda\int dr.r^2R_{21}R_{20} = \lambda\frac{\sqrt{3}}{2}$$

4. Use variation theorem to find the best possible value for the parameter α in $Ae^{-\alpha x^2}$ as the solution for the ground state of the 1-D harmonic oscillator.

Ans. $\psi_{\text{trial}} = Ae^{-\alpha x^2}$

Normalise : $\int\limits_{-\infty}^{\infty} A^2e^{-2\alpha x^2}dx = 1$ gives $A^2.\sqrt{\frac{\pi}{2\alpha}} = 1$ and $\therefore A = \left(\frac{2\alpha}{\pi}\right)^{\frac{1}{4}}$

Kinetic energy, $\langle T\rangle = \left\langle Ae^{-\alpha x^2}\left|-\frac{\hbar^2}{2\mu}\frac{d^2}{dx^2}\right|Ae^{-\alpha x^2}\right\rangle = -\frac{\hbar^2}{2\mu}.A^2\int\limits_{-\infty}^{\infty} e^{-\alpha x^2}\frac{d^2}{dx^2}\left(e^{-\alpha x^2}\right)dx$

$= -\frac{\hbar^2}{2\mu}\left[-2\alpha\int\limits_{-\infty}^{\infty} A^2e^{-2\alpha x^2}dx + 4\alpha^2\int\limits_{-\infty}^{\infty} A^2x^2e^{-2\alpha x^2}dx\right] = \frac{\hbar^2}{2\mu}\left[-2\alpha + 4\alpha^2.\sqrt{\frac{2\alpha}{\pi}}.\frac{1}{2}\sqrt{\frac{\pi}{(2\alpha)^3}}\right] = \frac{\alpha\hbar^2}{2\mu}$ (using $\int\limits_{-\infty}^{\infty} x^2e^{-ax^2}dx = \frac{1}{2}\sqrt{\frac{\pi}{a^3}}$)

Potential energy, $\langle V\rangle = \left\langle Ae^{-\alpha x^2}\left|\frac{1}{2}\mu\omega^2x^2\right|Ae^{-\alpha x^2}\right\rangle = \frac{1}{2}\mu\omega^2A^2\int\limits_{-\infty}^{\infty} x^2e^{-2\alpha x^2}dx = \frac{1}{2}\mu\omega^2.\frac{1}{4\alpha} = \frac{\mu\omega^2}{8\alpha}$

Total energy : $\langle H\rangle = \frac{\alpha\hbar^2}{2\mu} + \frac{\mu\omega^2}{8\alpha}$

differentiating for minimisation w.r.t. variation in α, we have, $\alpha = \frac{\mu\omega}{2\hbar}$

Notice : Putting this value of α back into the energy, we have the ground state energy $= \frac{\mu\omega}{2\hbar}.\frac{\hbar^2}{2\mu} + \frac{\mu\omega^2}{8}.\frac{2\hbar}{\mu\omega} = \frac{1}{2}\hbar\omega$

and $\psi = \left(\frac{2\frac{\mu\omega}{2\hbar}}{\pi}\right)^{\frac{1}{4}}.e^{-\frac{\mu\omega}{\hbar}x^2} = \left(\frac{\mu\omega}{\pi\hbar}\right)^{\frac{1}{4}}.e^{-\frac{\mu\omega}{2\hbar}x^2}$

Both these turn out to be exact expressions. Food for thought : Why/How did we get the exact results?

5. (a) What is the difference between a quantum state $|\psi\rangle = \sum\limits_i c_i|u_i\rangle$ and an ensemble of states $|u_i\rangle$ prepared as a classical statistical mixture?

(b) Define : density operator for an ensemble : $\hat{\rho} = \sum\limits_i c_i|u_i\rangle\langle u_i|$. How will you write the density operator for the 'pure' state in part (a).

(c) Show : (i) $\hat{\rho}$ is Hermitian.

(ii) $\mathcal{T}r(\rho) = 1$, where $\mathcal{T}r$ is the trace operator and ρ is the density matrix.

(iii) $\mathcal{T}r(\rho^2) = 1$

(iv) For a 'mixed' state (prepared as an ensemble), $\mathcal{T}r(\rho^2) < 1$

(v) The expectation value of an operator $\hat{A}$ can be calculated using $\mathcal{T}r(\rho A)$.

Ans. (a) The state $|\psi\rangle = \sum_i c_i |u_i\rangle$ is a 'pure', 'coherent' superposition of states $u_i\rangle$, each contributing in the proportion $|c_i|^2$, such that $\sum_i |c_i|^2 = 1$. The state $|\psi\rangle$ is known exactly, even though there is a probability of its being in the state $|u_i\rangle$ given by $|c_i|^2$.

An ensemble of states $|u_i\rangle$ prepared as a classical statistical mixture is a 'mixture' of pure states, each contributing in a certain ratio, say p_i, such that the state is $\sum_i p_i |u_i\rangle$.

(b) For the 'pure' state, density operator, $\hat{\rho} = |\psi\rangle\langle\psi|$.

(c) (i) $\hat{\rho}^\dagger = [|\psi\rangle\langle\psi|]^\dagger = |\psi\rangle\langle\psi| = \hat{\rho} \implies \hat{\rho}$ is Hermitian.

(ii) In the basis $|u_i\rangle$

The matrix elements $\rho_{ij} = \langle u_i|\hat{\rho}|u_j\rangle = \langle u_i|\psi\rangle\langle\psi|u_j\rangle = c_i^* c_j \langle u_i|u_j\rangle$

The trace is a sum over diagonal elements :

$\mathcal{T}r(\rho) = \mathcal{T}r(\langle\psi|\psi\rangle) = \sum_i |c_i|^2 \langle u_i|u_i\rangle = \sum_i |c_i|^2 = 1$ (assuming normalised basis)

(iii) $\hat{\rho}^2 = (|\psi\rangle\langle\psi|)\,(|\psi\rangle\langle\psi|) = |\psi\rangle\langle\psi| = \hat{\rho}$

$\therefore \mathcal{T}r(\rho^2) = \mathcal{T}r(\rho) = 1$

(iv) For a 'mixed' state, $\hat{\rho} = \left(\sum_k p_k |u_k\rangle\right)\left(\sum_l p_l \langle u_l|\right)$

and the matrix elements $\rho_{ij} = \langle u_i|\hat{\rho}|u_j\rangle = \langle u_i| \left(\sum_k p_k |u_k\rangle\right)\left(\sum_l p_l \langle u_l|\right)|u_j\rangle = p_i p_j$

$\therefore \mathcal{T}r(\rho) = \sum_i p_i^2 < 1$

(v) For the pure state, in the basis $|u_i\rangle$, $\mathcal{T}r(\rho A) = \langle u_i|\psi\rangle\langle\psi|\hat{A}|u_i\rangle = \sum_i |c_i|^2 A_{ii} = \langle\psi|\hat{A}|\psi\rangle$.